

GUIDE SPECIFICATION

Vertiv™ Liebert® APM2
60-300 kW and 60-600 kW; 400V

Vertiv™ Liebert® APM2

60-300kW and 60-600 kW; 400V

GUIDE SPECIFICATIONS

1.0 GENERAL

1.1 SUMMARY

The specifications mentioned in this document describe the requirements for a Modular, Scalable, & Digital Uninterruptible Power Supply (UPS) consisting of one or more single power module units connected in parallel inside a standard frame without the need for either a centralized system controller or an external centralized maintenance bypass switch. The UPS shall automatically maintain AC power within specified tolerances to the critical load, without interruption (for specified duration as per battery run time), during failure or deterioration of the mains power supply. The UPS system shall be expandable by inserting additional modules of the same rating, to provide for module redundancy or load growth requirements. The manufacturer shall design and furnish all materials and equipment to be fully compatible with electrical, environmental and space conditions at the site. The UPS includes all equipment to properly interface the AC power source to the intended load and will be designed for unattended operation.

1.2 STANDARDS

The Modular UPS system along with the associated equipment and components will be manufactured in accordance with the following applicable standards:

General safety requirements for UPS	IEC 62040-1
EMC requirements for UPS	IEC 62040-2
Method of specifying the performance and test requirements of UPS	VFI-SS-111
Environmental aspects - requirements and reporting	EN62040-4/IEC62040-4/AS62040-4 (VFI SS 111)

The UPS is CE marked in accordance with the requirements set out in Directive 2014/35/EU Directive 2014/30/EU, Directive 2011/65/EU (with amendment Directive (EU)2015/863).

The Quality System for the engineering and manufacturing facility certificated to conform to Quality System Standard ISO 9001 for the design and manufacture of power protection systems for computers and other sensitive electronics.

1.3 SYSTEM DESCRIPTION

1.3.1 Design Requirements

- A. For non-redundant operation (applicable, not applicable), the Complete integrated UPS system will be _____ sized to provide a minimum of ____ kVA / ____ kW quality power output at ____°C ambient in N

- B.** For redundant operation (applicable, not applicable), the complete integrated UPS system will be___ sized to provide a minimum of ___ kVA / ___kW quality power output at ___°C ambient in N+__ (1, 2, 3)

Load voltage and bypass line voltage shall be 400VAC, three-phase, four-wire plus ground. Input voltage shall be 400VAC, three-phase, four-wire plus ground. The AC input source and bypass input source shall each be a solidly grounded wye service. It will be able to supply all required power to full rated output kVA loads with power factor from 0.5 lagging to 0.5 leading without any deration.

The battery will support the UPS a capacity of ___ kW load for at least ___ minutes at 25°C at startup.

1.3.2 Modes of Operation

The UPS shall operate as an on-line reverse transfer system in the following modes:

- A. Normal:** The UPS inverter continuously supplies the critical AC load. The rectifier draws power from the commercial AC source and converts it into DC power for the inverter and the battery charger. The battery charger maintains the battery in a fully charged and optimum operational condition. The inverter converts the DC power into clean and regulated AC power which is supplied to the critical load (conditioned line).
- B. ECO Mode:** The critical AC load shall be continuously powered by the bypass with the inverter available to power the load if the bypass source voltage or frequency exceeds adjustable parameters of power quality. This feature shall also be available in parallel operation.
- C. Dynamic Online Mode:** If this mode is selected, all power switches (configured internally or externally) and the battery switches are closed except for the maintenance bypass switch, and the system prefers to put the load on the bypass, to achieve the aim of energy-saving. When the load power is fed by bypass supply, the inverter is in the mode of power quality compensation for the bypass voltage. When the voltage of the bypass supply is beyond the pre-defined and adjustable limits, the system will transfer to the inverter output. In this mode, the system can normally charge the battery. This feature shall also be available in parallel operation.
- D. Battery:** Upon failure of utility AC power, the critical load shall be powered by the inverter, which, without any switching, shall obtain its power from the battery plant via the DC-DC converter. There shall be no interruption in power to the critical load upon failure or restoration of the utility AC source.
- E. Recharge:** Upon restoration of the utility AC source, the rectifier shall supply power to the output inverter and to the DC-DC converter, which shall simultaneously recharge the batteries. This shall be an automatic function and shall cause no interruption to the critical load.
- F. Bypass:** If the UPS must be taken out of service, the static transfer switch shall transfer the load to the bypass source. The transfer process shall cause no interruption in power to the critical load. An optional external wrap-around maintenance bypass shall be used to ensure full isolation of the unit for the service of internal components while providing safety from arc flash.
- G. Off-Battery:** If the battery is taken out of service, it shall be disconnected from the DC-DC converter by means of an external disconnect circuit breaker. The UPS shall continue to function and meet all the specified steady-state performance criteria, except for the power outage backup time capability. If multiple battery strings are used, each string shall be capable of being electrically isolated for safety during maintenance.
- H. Parallel Operation:** The UPS shall have the vertical & horizontal scalability option for redundancy or capacity expansion
- 1. Vertical Scalability:** UPS shall be capable to accommodate additional power modules within the same rack (limited to its maximum frame capacity) online without dropping the connected load. It shall be designed based on discrete control logic platform i.e. each power & static switch module shall have its own intelligent DSP controller to avoid single point of failure.

2. Horizontal Scalability: UPS shall have option to connect the two UPSs in parallel configuration for redundancy or capacity expansion. Under normal operating conditions, the power delivered to the load will be equally shared between number of UPS power modules connected to the parallel bus with a tolerance of 5%. In the event of an overload, the system will transfer to the bypass source.

- I. Online Maintenance:** If the UPS needs to be maintained or repaired, all UPS modules & static switch module in the system should be replaceable online without shutting the system down. Additionally, UPS should also be equipped with maintenance bypass for full system maintenance.
- J. Common Battery:** UPS shall be capable to support common battery configuration when multiple UPSs are connected in parallel. In this mode, each UPS can use the same battery to feed the required load.
- K. Regen Mode:** The UPS rack system shall have the ability to perform self-test for full rated capacity without using any external load banks. In this mode, UPS rectifier, inverter and static bypass shall be tested up to full load capacity without any failure. Power consumption in this mode shall only be full load losses of UPS. This mode shall also be available in parallel UPS configuration.

1.3.3 Performance Requirements

The solid-state power components, magnetics, electronic devices, and overcurrent protection devices shall operate within the manufacturer's recommended temperature when the UPS is operating at 100% critical load and maintain battery charging under either of the following conditions:

- Any altitude within the specified operating range ≤ 1500 m elevation.
- Any ambient temperature within the specified operating range of 0°C to 50°C .

1.3.4 Input

- A. Voltage:** Input voltage specifications of the UPS shall be
 - Rectifier AC Input: 380/400/415V, three-phase, four-wire-plus-ground
 - Bypass AC Input: 380/400/415V, three-phase, four-wire-plus-ground
- B. Voltage Range:** 305-478 V at full at full load; 228 – 478 V at 75% derated load conditions without battery discharge
- C. Frequency Range:** 40 - 70Hz
- D. Power Factor:** From 50%-100% load, input power factor min is 0.99
- E. Total Current Harmonic Distortion :** The Total Harmonic Distortion of input current introduced by the rectifier will be less than or equal to 10% for 25% linear load, 3% for 100% linear load and 5% for 100% nonlinear load.
- F. Power walk-in period:** 5 seconds default (selectable 1 through 30 seconds)

1.3.5 AC Output

- A. AC Output:** 380/400/415 V, three-phase and sharing neutral with bypass input
- B. Load Rating:** 100% of load rating for any load from 0.5 lagging to 0.5 leading
- C. Voltage Stability:** $\pm 1\%$ RMS average for a 100% balanced three-phase load; $\pm 3\%$ RMS average for a 100% unbalanced load
- D. Voltage Adjustment Range:** $\pm 5\%$ for line drop compensation adjustable by factory service personnel
- E. Frequency Stability:** Synchronized with internal clock: $\pm 0.05\%$
Synchronized with bypass: $\pm 0.25\%$

F. System Efficiency: Defined as output kW/input kW at rated lagging load power factor; and not less than the values listed below

In Double conversion mode: Up to 97.5%

In Dynamic online mode: Up to 98.9 ($\pm 0.2\%$)

In Eco mode: Up to 99%

G. Phase Imbalance:

- i. Balanced loads $120^\circ \pm 1^\circ$
- ii. 100% unbalanced loads $120^\circ \pm 1.5^\circ$

H. Voltage Transients (average of all three phases):

- i. 0-100% or 100-0%
Response Meets IEC 62040-3: 2010 Figure 2 Curve 1, Class 1
Meets ITIC and CBEMA Curve Requirements
- ii. 10-100% or 100-10%
Transient Voltage Deviation, RMS 5%

I. Overload at Full Output Voltage:

- i. <105% continuous
- ii. 105%-110% for 1 hour
- iii. 110% - 125% for 10 minutes
- iv. 125% - 140% from 10 to 1minute smoothly
- v. 140%-150% for 1 minute
- vi. >150% for 200 ms

1.3.6 Grounding

The UPS chassis shall have an equipment ground terminal.

1.4 ENVIRONMENTAL CONDITIONS

The UPS shall be able to withstand the following environmental conditions without damage or degradation of operating characteristics:

A. Operating Ambient Temperature

- UPS: 0 °C to 40 °C full performance, 40 °C to 50 °C with automatic derating,
- Battery Temperature compensation.: -3.0 (selectable from 0 to -5.0 around 25°C)

B. Storage/Transport Ambient Temperature

- Storage: -25 °C ~ +55 °C; transportation: -40 °C ~ +70 °C

C. Relative Humidity

- 0 to 95%, non-condensing

D. Altitude

- ≤ 1500 m; derate power by 1% per 100 m between 1500 m and 3000 m

E. Audible Noise Level

- <65 dBA measured 1m from the surface of the unit for 300 kVA
- <70 dBA measured 1m from the surface of the unit for 600 kVA

1.5 WARRANTY

1.5.1 UPS Warranty

The UPS manufacturer shall warrant the unit against defects in workmanship and materials for 12 months after initial startup or 18 months after the shipping date, whichever comes first.

1.5.2 Warranty – End User

Warranties associated with items not manufactured by the UPS supplier but included as part of the system shall be passed through to the end user.

1.6 QUALITY ASSURANCE

1.6.1 Manufacturer's Qualifications

A minimum of 30 years' experience in the design, manufacture, and testing of solid-state UPS systems shall be required. The quality system for the engineering and manufacturing facility shall be certified to conform to Quality System Standard ISO 9001 for the design and manufacture of power protection systems for computers and other sensitive electronics.

1.6.2 Factory Testing

- Before shipment, the manufacturer shall fully and completely test the UPS unit to ensure compliance with the specification.
- The UPS unit shall be tested at the system-specified capacity. Testing shall be done using load banks at part-load and the full kW rating of the unit.
- Operational discharge and recharge tests shall be conducted to ensure performance.
- System operations such as startup, shutdown, and transfers shall be demonstrated.
- A certified copy of the test results shall be available for each system as indicated on the order.

2.0 PRODUCT

2.1 FABRICATION

2.1.1 Materials

All materials of the UPS shall be new, of current manufacture, high grade and shall not have been in prior service except as required during factory testing. All active electronic devices shall be solid-state. All power semiconductors shall be sealed. Control logic and fuses shall be physically isolated from power train components to ensure operational safety and to protect from heat. Minimum estimated life of the major components used in the UPS (includes fan & bus capacitors) shall not be less than 7 years.

2.1.2 UPS Internal Wiring

Wiring practices, materials, and coding shall be in accordance with the requirements of the National Electrical Code and applicable local codes and standards. All bolted connections of bus bars, lugs, and cables shall be in accordance with requirements of the National Electric Code and other applicable standards. All electrical power connections shall be torqued to the required value and marked with a visual indicator.

2.1.3 Field Wiring

All field wiring power connections shall be to tin-plated copper bus bars for connection integrity. Bus bars shall have adequate space to allow two-hole, long-barrel, and compression type lugs forming a permanent connection between field wiring and field-installed lugs.

Provisions shall be made in the cabinets to permit installation of input, output, and external control cabling using raceway or conduit. Provision shall be made for top and bottom access to input, output, bypass, and DC connections as per the technical specifications. In conformance with the NEC, connection cabinets shall provide for adequate wire bend radius.

2.1.4 Construction and Mounting

The UPS shall be housed in an IP20 enclosure, designed for floor mounting. The UPS shall be structurally adequate and have provisions for forklift handling. Maximum cabinet height shall be 2 meters for all UPS range.

2.1.5 Cooling

Forced air cooling shall be provided to ensure that all components are operated well within temperature ratings. Airflow shall be controlled according to load demand. If one of the cooling fans experiences a fault, the UPS shall be immediately notified of the condition via the user interface and through remote monitoring services.

2.2 Equipment

2.2.1 UPS System

The UPS system shall consist of an IGBT power factor-corrected rectifier, DC-DC converter and three-phase, transformer-free T-type inverter, bypass static transfer switch, bypass synchronizing circuitry, protective devices, and accessories as specified. The specified system shall also include a battery disconnect breaker and battery system.

2.2.2 Surge Protection

The UPS shall have built-in protection against surges, sags, and overcurrent from the AC source. The protection shall meet the requirements specified in standards EN 62040-2:2006 & IEC/EN 61000-4-5:2006.

2.2.3 Output Protection

The UPS shall be protected against sudden changes in output load and short circuits at the output terminals. The UPS shall have built-in protection against permanent damage to itself and the connected load for all predictable types of malfunctions. Fast-acting, current-limiting devices shall be used to protect against cascading failure of solid-state devices. Internal UPS malfunctions shall cause the module to trip off-line with minimum damage to the module and provide maximum information to maintenance personnel regarding the reason for tripping off-line. The load shall be automatically transferred to the bypass line uninterrupted for an internal power modules malfunction. The status of protective devices shall be indicated on a graphic display screen on the front of the unit.

2.3 Components

2.3.1 Rectifier

The term rectifier shall denote the solid-state equipment and controls necessary to convert alternating current to regulated direct current to supply the inverter and charge the battery. The DC output of the rectifier shall meet the input requirements of the inverter without the battery being connected.

A. Input Current Harmonic Distortion

The rectifier shall actively control and reduce input current distortion over the full operating range of the UPS without the need for an additional passive input filter. Input current THD shall be less than 3% at rated linear loads and nominal voltage in double-conversion mode.

B. Dynamic Current Input Limit Reduction

The rectifier, in conjunction with the other UPS controls and circuitry, shall adjust the current demanded for battery charging as a function of UPS wattage load and input voltage level.

2.3.2 DC-DC Converter

The term DC-DC converter shall denote the equipment and controls to regulate the output of the rectifier to the levels appropriate for charging the battery and to boost the battery voltage to the level required to operate the inverter. The DC-DC converter shall be solid-state, capable of providing rated output power and, for increased performance, shall be a pulse width-modulated design and shall utilize insulated gate bipolar transistors (IGBTs). The DC-DC converter shall control charging of the battery. The AC ripple voltage of the charger DC shall not exceed 1.4% RMS of battery voltage

A. Battery Recharge

The rectifier/charger shall be capable of supplying an additional 15% power for recharging a fully discharged DC stored energy source for charging current with UPS at full load and input voltage at nominal. After the stored energy source is recharged, the rectifier/charger shall maintain the source at full charge until the next emergency operation.

B. Battery Equalize Charge

A manually initiated equalize charge feature shall be provided to apply an equalize voltage to the battery. The duration of equalize charge time shall be adjustable from 8 to 30 hours. A method shall be available to deactivate this feature for valve regulated battery systems.

C. Stop Battery Charging Function

Battery charging may be stopped by a shunt trip of the battery cabinet breaker when over temperature is sensed in the battery cabinet, on generator or when environmental contact is closed.

D. Overvoltage Protection

There shall be DC overvoltage protection so that if the DC voltage rises to the pre-set limit, the UPS shall shut down automatically and initiate an uninterrupted load transfer to bypass or shall disconnect the battery via the DC breaker(s) in the battery string.

E. Temperature-Compensated Charging

The UPS shall adjust the battery charging voltage based on the battery temperature reported from external battery temperature sensors. When multiple sensors are used, the voltage shall be based on the average temperature measured. Excessive difference in the temperature measurements shall be reported and the charging voltage adjusted to protect the batteries from excessive current.

F. Battery Load Testing

The UPS shall be capable of performing battery load testing under operator supervision. The UPS shall be capable of testing the battery without risk of disrupting power to the critical load. The UPS shall display and log the result of the test with appropriate alarms if reduced battery capacity or battery failure is detected.

2.3.3 Inverter

The term inverter shall denote the equipment and controls to convert direct current from the rectifier or battery via the DC-DC converter to precise alternating current to power the load. The inverter shall be solid-state, capable of providing rated output power and, for increased performance; the inverter shall be a pulse-width-modulated design and shall utilize insulated gate bipolar transistors (IGBTs).

A. Overload Capability

The inverter shall be able to sustain an overload across its output terminals while supplying full rated voltage of up to 150% for 60 seconds. The inverter shall be capable of at least 250% current for short-circuit conditions including phase-to-phase, phase-to-ground and three-phase faults. After the fault is removed, the UPS shall return to normal operation without damage. If the short circuit is sustained, the load shall be transferred to the bypass source and the inverter shall disconnect automatically from the critical load bus.

B. Output Frequency

The inverter shall track the bypass continuously, providing the bypass source maintains a frequency of Nominal $\pm 5\%$ Hz

C. Phase-to-Phase Balance

The inverter shall provide a phase-to-phase voltage displacement of no worse than $\pm 1.5^\circ$ with a 100% unbalanced load.

D. Inverter Fault Sensing and Isolation

The UPS shall be provided with a means to detect a malfunctioning inverter and isolate it from the critical load bus to prevent disturbance of the critical load voltage beyond the specified limits.

E. Battery Protection

The inverter shall be provided with monitoring and control circuits to protect the battery system from damage due to excessive discharge. Inverter shutdown shall be initiated when the battery voltage has reached the end of discharge voltage. The battery end-of-discharge voltage shall be calculated and automatically adjusted for partial load conditions to allow extended operation without damaging the battery. Automatic shutdown based on discharge time shall not be acceptable.

2.3.4 Inverter Bypass Operation

When maintenance is required or when the inverter cannot maintain voltage to the load due to sustained overload or malfunction, a bypass circuit shall be provided to isolate the inverter output from the load and provide a path for power directly from an alternate AC (bypass) source. The UPS control system shall constantly monitor the availability of the inverter bypass circuit to perform a transfer. The inverter bypass circuit shall consist of a continuous duty bypass static switch and an overcurrent protection device to isolate the static bypass switch from the bypass utility source. The bypass static switch shall denote the solid-state device incorporating SCRs (silicon-controlled rectifiers) that can automatically and instantaneously connect the alternate AC source to the load.

A. Static Bypass Switch Rating

The static bypass switch shall be rated for continuous duty operation at full rated load for highest reliability without the use of mechanical devices as used with a momentary rated device.

B. Manual Load Transfers

A manual load transfer between the inverter output and the alternate AC source shall be initiated from the control panel. Manually initiated transfers shall be make-before-break, utilizing the inverter and the bypass static switch.

C. Automatic Load Transfers

An automatic load transfer between the inverter output and the alternate AC source shall be initiated if an overload condition is sustained for a period in excess of the inverter output capability or due to a malfunction that would affect the output voltage. Transfers caused by overloads shall initiate an automatic retransfer of the load to the inverter only after the load has returned to a level within the rating of the inverter source and the alarm has been acknowledged.

D. Momentary Overloads

In the event of a load current inrush or branch load circuit fault in excess of the inverter rating, the bypass static switch shall connect the alternate AC source to the load for at least 100 milliseconds, allowing up to >400% of the normal rated output current to flow. Output voltage shall be sustained to the extent the alternate AC source capacity permits. If the overload condition is removed before the end of the 100 milliseconds period, the bypass static switch shall turn Off and the load shall remain on inverter power. If the overload remains, then a transfer to the alternate AC source is to be completed.

E. Back-Feed Protection

As required by IEC/EN 62040-1, the static transfer switch shall not back-feed UPS power to the bypass distribution system while the UPS is operating on battery during a bypass power outage. The purpose of this requirement is to prevent the risk of electrical shock on the distribution system when the normal source of power is disconnected or has failed. If a shorted SCR is detected, the static transfer switch shall be isolated by an external contactor based circuit breaker and an alarm message shall be annunciated at the UPS control panel. The load shall remain on conditioned and protected power after detection of a shorted SCR and isolation of the bypass static switch.

F. ECO-Mode

When selected, this mode of operation shall transfer the load to the bypass source and maintain it there as long as the bypass source frequency, slew rate, and voltage are within the adjusted operating parameters. While in this mode, the inverter shall remain operating to demonstrate the ability to instantaneously assume the load without interrupting the output voltage. Should the bypass source go outside the adjusted limits, the bypass static switch shall turn Off, isolating the load from the bypass while the inverter assumes the full critical load. The load shall be transferred from the bypass source to the inverter while maintaining the output voltage within the ITIC and CBEMA curves. There is provision to activate this function via touch screen display.

2.3.5 Dynamic Online Mode

This functioning mode allows significant energy savings by operating with a typical efficiency above 98.8% while providing power conditioning to the load.

Normal (VI)

The operating mode will depend on the quality of the mains supply in the short-term past and on the electrical characteristic of the load. If the line quality remains within permitted tolerance parameters and the load needs power conditioning, (THDi, THDv, PF) the power interface will provide continuous supply to the critical AC load while the inverter operates as series and parallel active filter. The IGBT inverter will be able to compensate the power factor of the load and the current harmonic distortion guaranteeing optimum power conditioning to the load while maintaining the highest level of efficiency.

Fast Transfer to VFI

Emergency (due to mains supply failure or variance beyond tolerance limits) if the bypass mains supply varies beyond tolerance levels (adjustable using the software) that cannot be compensated through the active filter, the load will be transferred from the direct line to the conditioned line, with Class 1 output performance. The load is powered from the mains via the rectifier and inverter, (provided the input mains remains within the tolerances stated in technical specifications). If the input mains fall below the lowest limit, the batteries will be used to power the load via the inverter.

Return to VI

When the mains supply returns within tolerance limits, the UPS will continue to supply the load via the conditioned line for a period of time dependant on the direct line failure rate (the conditioned line draws power from the mains not the battery). When the direct line has stabilized, the UPS returns to normal VI operation. The battery charger automatically begins to recharge the battery so that maximum autonomy is guaranteed in the shortest possible time.

2.3.6 Display and Controls

A. UPS Control Panel

Each UPS module shall be equipped with a 9-inch touch screen, graphical LCD display. That will automatically provide all information relating to the current status of the UPS as well as display metered values. The display shall be menu-driven, permitting the user to easily navigate through operator screens. The flexibility shall also be provided to change the display arrangement and information displayed as per the user requirement.

B. Logic

UPS system logic and control programming shall reside in a microprocessor-based control system with nonvolatile flash memory. Rectifier, inverter, and system control logic shall utilize high-speed digital signal processors (DSPs). CAN bus shall be used to communicate between the logic and the User Interface as well as the options. Switches, contacts, and relays shall be used only to signal the logic system as to the status of mechanical devices or to signal user control inputs. Customer external signals shall be isolated from the UPS logic by relays or optical isolation.

C. Metered Values

A microprocessor shall control the display and memory functions of the monitoring system. All three phases of three-phase parameters shall be displayed simultaneously. All voltage and current parameters shall be monitored using true RMS measurements for accuracy to $\pm 2\%$ of AC voltage, $\pm 5\%$ AC current, $\pm 0.2\%$ AC frequency, $\pm 5\%$ battery voltage, and $\pm 5\%$ battery current. The following parameters shall be displayed:

- Input voltage, line-to-line
- Input current per phase
- Input frequency
- Input Power factor
- Battery voltage
- Battery charging/discharging current
- Output voltage, line-to-line
- Output frequency
- Bypass input voltage, line-to-line
- Bypass input frequency
- Load current
- Load real power (kW), total and percentage
- Load apparent power (kVA), total and percentage
- Load percentage of capacity
- Battery temperature, each battery string
- Battery state of charge
- Real time efficiency curve

D. Power Flow Indications

A power flow diagram shall graphically depict whether the load is being supplied from the inverter, bypass or battery and shall provide, on the same screen, the status of the following components:

- AC Input Circuit Breaker (optional)
- Battery Circuit Breaker, each breaker connection of complete battery complement, complete disconnection and partial connection (one or more, but not all breakers open.)
- Maintenance Bypass Status

E. Main Display Screen

The following UPS status messages shall be displayed:

- Rectifier (Off / Soft Start / Main Input On / Battery Input On)
- Input Supply (Normal Mode / Battery Mode / All Off)
- Battery Self-Test (True / False)
- Input Disconnect (Open / Closed)
- EPO (True / False)
- Charger (On / Off)
- Output Disconnect (Open / Closed)
- Maint. Disconnect (Open / Closed)

- Bypass Disconnect (Open / Closed)
- Inverter (Off / Soft Start / On)
- Bypass (Normal / Unable to Trace / Abnormal)
- Output Supply (All Off / Bypass Mode / Inverter Mode / Output Disable)
- Inverter On (Enable / Disable)

F. HMI Control Buttons

Buttons shall be provided to start and stop the inverter. A pop-up message requesting confirmation shall be displayed whenever a command is initiated that would change the status of the UPS. Other buttons shall be provided to reset faults and silence the alarm buzzer.

G. Event Log

This menu item displays the list of events that have occurred recently while the UPS was in operation. The Event Log stores min of 1000 events, with the oldest events being overwritten first if the log's capacity is reached.

H. Battery Status Indicator

A battery status indicator displays DC alarm conditions, temperature, battery state of charge, the present battery voltage, total discharge time, status of last battery test, and battery time remaining during discharge.

The UPS provides the operator with controls to perform the following functions:

- Configure and manage manual battery test.
- Modify test duration and minimum voltage
- Start battery test
- Monitor test status and progression
- Stop battery test
- Battery test status

I. Alarms

The following alarm messages shall be displayed:

- | | |
|--------------------------|--|
| • Mains Voltage Abnormal | • Transfer Time-Out |
| • Mains Under voltage | • Load Sharing Fault |
| • Mains Freq. Abnormal | • Bypass Over Current |
| • Charger Fault | • DC Cap Life Pre-warning |
| • Battery Reversed | • DC Cap Life Warning |
| • No Battery | • Fan Life Pre-warning |
| • Parallel Comm. Fail | • Fan Life Warning |
| • Bypass Unable to Track | • Intelligent Parallel Mode Active |
| • Bypass Abnormal | • Input Neutral Lost |
| • Inverter Asynchronous | • Loss of Redundancy |
| • Fan Fault | • Multi Module System Inverter Manual ON |
| • Control Power Fail | |
| • Unit Overload | |
| • System Overload | |
| • Bypass Phase Reversed | |

J. Controls

System-level control functions are:

- Start Inverter (and transfer to inverter)
- Stop Inverter (after transferring to bypass)
- Startup Screen
- Battery Test Set Point Adjustment
- Configure Manual Battery Test
- Initiate Manual Battery Test
- System Settings (Time, Date, Language, LCD Brightness, Password, Audio Level)
- Alarm Silence Command
- Fault Reset Command
- Energy Saving Modes
- Warning & Critical Thresholds of Load and Battery

K. Manual Procedures

- Load Transfers: HMI buttons (START INVERTER, STOP INVERTER) provides the means for the user to transfer the load to bypass and back on UPS.

L. Self-Diagnostics

- Event Log File - The control system shall maintain a log of the event conditions that have occurred during system operation. Each log contains the event name, event time/date stamp and a set/clear indicator.
- Fault waveform capturing- Records 3 cycles data of voltage/current of rectifier/inverter/bypass when fault triggers; In 3 cycle of recorded data, 2 cycle of data shall be captured before trigger point, 1 cycle of data shall be captured after fault trigger point.

2.3.7 Remote Monitoring and Integration Capabilities

24 X 7 remote monitoring services: The UPS manufacturer shall provide as an optional service, which provides 24x7 continuous monitoring of events and parametric data, event and data analysis reports, and dispatch of factory trained field service personnel. The UPS is able to initiate periodic and critical event-driven communication with a remote service center to transfer event and parametric data for analysis and action. The remote service centers are staffed with factory-trained service personnel who can receive, analyzing, and interpreting the communicated events and data. In case of any anomalies, the remote service center personnel will dispatch factory-trained field service personnel with necessary tools & accessories to the location of the UPS to fix the problem.

A. **Communication Cards:** The UPS can be equipped with following communication card(s) including:

- **Optional Communication card-** This card provides Web access, environmental sensor data, and third-party customer protocols for the UPS and manage a wide range of operating parameters, sending data over ethernet networks via secure HTTPS protocol and alarms and notifications via SNMP traps or Modbus protocol. It is also capable to integrate with any existing building management system.
- **Optional Software-** Provided to support the monitoring of multiple no of UPS systems at single console platform
- **Relay card:** Dry contact outputs are provided for Summary Alarm, Bypass Active, Low Battery, and AC Input Failure.

- B. **Customer Input Contacts:** The UPS shall have four discrete input contacts available for the input and display of customer-provided alarm points or to initiate a pre-assigned UPS operation. Each input can be signaled by an isolated, external, normally open contact.

When an assembly is selected as a pre-assigned UPS operation, the following actions should be initiated:

- On Generator— Provides selectable choices to enable or disable battery charging and enable or disable ECO Mode operation while on generator.
- Transfer to Bypass—Manual command to transfer from inverter operation to static bypass operation.
- Fast Power Off—Emergency Module Off (EPO) command to stop UPS operation.
- Acknowledge Fault—Acknowledge a UPS alarm condition and present faults will be reset.
- Bypass/Inverter Off—Emergency Power Off (EPO) command to stop UPS operation.
- External Maintenance Bypass Breaker (MBB) status (open or closed).

2.3.8 Battery Disconnect Breaker

The battery cabinet shall have a properly rated circuit breaker to isolate it from the UPS. This breaker will be in a separate enclosure or in a matching UPS/battery cabinet. When this breaker is open, there shall be no battery voltage in the UPS enclosure. The UPS will be automatically disconnected from the battery by a shunt trip of the battery cabinet breaker when signaled by other control functions.

2.3.9 Optional Accessories and Features

A. Load Bus Sync

The Load Bus Sync (LBS) enables two independent single-module UPS units to stay in sync when operating on battery or unsynchronized input sources. The LBS determines the master and slave relationship between UPS units. The LBS will be installed within each single-module UPS.

B. Seismic Anchorage Kits

Seismic anchorage kits will be provided with the UPS unit, and if included the optional matching battery cabinet, for use in seismic restraint as required for IBC 2012 or OSHPD certification.

3.0 EXECUTION

3.1 FIELD QUALITY CONTROL

The following inspections and test procedures shall be performed by factory-trained field service personnel during the UPS startup.

A. Visual Inspection

- Inspect equipment for signs of damage.
- Verify installation per drawings supplied with installation manuals or submittal package.
- Inspect cabinets for foreign objects.
- Verify that neutral and ground conductors are properly sized and configured per OEM requirements as noted in OEM drawings supplied with installation manuals or submittal package.
- Inspect each battery jar for proper polarity.
- Verify that all printed circuit boards are configured properly.

B. Mechanical Inspection

- Check all control wiring connections for tightness.
- Check all power wiring connections for tightness.
- Check all terminal screws, nuts, and/or spade lugs for tightness.

C. Electrical Inspection

- Check all fuses for continuity.
- Confirm input and bypass voltage and phase rotation are correct.
- Verify control transformer connections are correct for voltages being used.
- Ensure connection and voltage of the battery string(s).

3.2 UNIT STARTUP

- Energize control power.
- Perform control/logic checks and adjust to meet OEM specification.
- Verify DC float and equalize voltage levels.
- Verify DC voltage clamp and overvoltage shutdown levels.
- Verify battery discharge, low battery warning, and low battery shutdown levels.
- Verify fuse monitor alarms and system shutdown.
- Verify inverter voltages and regulation circuits.
- Verify inverter/bypass sync circuits and set overlap time.
- Perform manual transfers and returns.
- Simulate utility outage at no load.
- Verify proper recharge.

3.3 MANUFACTURER'S FIELD SERVICE

A. Service Personnel

The UPS manufacturer shall directly employ a nationwide service organization, consisting of factory-trained field service personnel dedicated to the startup and maintenance of UPS and power equipment.

The manufacturer shall provide a national dispatch center to coordinate field service personnel schedules. One toll-free number shall reach a qualified support person 24 hour a day, 7 days a week and 365 days a year. If emergency service is required, on-site response time shall be 4 hours or less within 150 miles of a Services center.

Two local customer engineers shall be assigned to the site with a regional office as a backup. Escalation procedures shall be in place to notify Power technical support if a site is not functioning within 24 hours.

B. Replacement Parts Stocking

Parts shall be available through an extensive network to ensure round-the-clock parts availability throughout the country.

Spare parts shall be stocked by local field service personnel with backup available from national parts centers and the manufacturing location. A Customer Support Parts Coordinator shall be on call 24 hours a day, 7 days a week, 365 days a year for immediate parts availability.

C. Maintenance Contracts

A complete offering of preventive and full-service maintenance contracts for both the UPS system and battery system shall be available.

UNINTERRUPTIBLE POWER SYSTEM TECHNICAL DATA:

	Make			
	Model			
A	General Characteristics			Suppliers Data
1	Continuous duty three phase double conversion uninterruptible power system (UPS); The UPS shall utilize a standard 3U size rack mountable power module & scalable array architecture			
2	Hot swappable power module should have its own controller and contains a full rated rectifier, full rated inverter, and battery charging circuit.			
3	UPS comprises a user replaceable centralized continuous duty hot swappable bypass static switch module			
4	Standard configuration of UPS footprint shall not exceed for 300kW andfor 600 kW			
5	All the conformal coated PCBs shall be used in the modules			
6	External battery circuit breaker with necessary shunt tripping shall be provided for external battery banks			
B	System Characteristics			
	Item	Unit	Specifications	
	Dimensions, W × D × H	mm	For 300 kVA: Full Switch Assembly– 600 x 890 x 2000 mm For 600 kVA: Compact Version- 600 x 990x 2000 mm Full Switch Assembly- 1200 x 990 x 2000 mm	
	Weight	kg	For 300 kVA: Full Switch Assembly– 285 kg For 600 kVA: Compact Version- 510 kg Full Switch Assembly- 830 kg	
	Noise within 1m	dBA	For 300 kVA-Not more than 65 For 600 kVA- Not more than 70	
	Operating temperature	°C	0 °C to 40 °C full performance, 40 °C to 50 °C with automatic derating	
	Relative humidity	%RH	0 to 95%, non-condensing	
	Efficiency @ Double conversion mode at 400 VAC	%	Up to 97.5%	
	Efficiency @ Eco mode	%	Up to 99%	
	Efficiency @ dynamic online mode	%	Up to 98.9 (±0. 2%)	

	Protection degree, IEC (60529)	N/A	IP20	
C	Input Characteristics			
	Rated AC input voltage	Vac	380/400/415 V (3-phase 4-wire (+PE) TN/TT/IT power distribution system)	
	Input voltage tolerance	Vac	305-478 at full at full load; 228 - 478 at 75% derated load conditions without battery discharge	
	Frequency	Hz	50/60 (tolerance: 40~70)	
	Power factor	kW/kVA	0.99 (50-100% Load)	
	Maximum Input current	A rated (maximum)		
	Harmonic current distortion	THDI%	<=3 with linear load at full load operation	
	Power walk-in duration	s	5 seconds default (selectable 1 through 30 seconds)	
D	Intermediate DC Circuit			
	Battery bus voltage	Vdc		
	Total Battery Capacity (as per above backup time)	AH	40 to 50 block of 12V at full load, 38 block derated to 95%, 36 block derated to 90%, 34 block derated to 85%, 32 block derated to 80%, 30 block derated to 75%)	
	Max. battery charging current	A	For 300 kVA-600 For 600 kVA- 1200	
	Charging Mode		Constant current and constant voltage	
	Float voltage	V/cell (VRLA)	2.27V/cell (selectable from 2.2V/cell to 2.3V/cell)	
	Temperature compensation	mV/°C/cell	-3.0 (selectable from 0 to -5.0 around 25°C)	
	Ripple voltage	% V float	≤1.414	
	Ripple current	% C10	≤5	
E	UPS Output characteristics			
	Rated AC voltage	Vac	380/400/415 (three-phase four-wire, with neutral reference to the bypass neutral)	
	Frequency	Hz	50/60	
	Overload	%	≤ 105% Continuous	
			105% to 125% for 10 min	
			125% to 150% for 1 min	
			>150% for 200 ms	
	Non-linear load capability	%	100%	

	Steady state voltage stability	%	±1 (balanced load), ±3 (100% unbalanced load)	
	Transient voltage response	%	±5	
	Total harmonic voltage	%	<1 % (100%linear load), <3% (100%non-linear load)	
	Synchronization window	Hz	Upper limit: 0.5Hz, 1Hz, 2Hz, 3Hz, +10%; Default: +10%. Lower limit: -0.5Hz, -1Hz, -2Hz, -3Hz, -10%; Default: -10%	
	Slew rate (max change rate of synchronization frequency)	Hz/s	0.6; setting range: 0.1 ~ 3	
	Transfer time	ms	Frequency converter mode: transfer without interruption 0ms; transfer with interruption < =5 ms; ECO mode: 4 ms; Dynamic online mode: 0 ms.	
F	Bypass Input Characteristics			
	Rated AC voltage	Vac	380/400/415, three-phase four-wire, sharing neutral with the rectifier input and providing neutral reference to the output	
	Overload	%	110% for continuous	
			125% for 10 min	
			150% for 60 sec	
			>150% for 200 ms.	
	Frequency	Hz	50/60	
	Bypass voltage tolerance	%Vac	Upper limit: +10% Vac, +15% Vac, or +20% Vac Default: +15% Vac	
			Lower limit: -10% Vac, -20% Vac, -30% Vac, or -40% Vac Default: -20% Vac	
	Bypass frequency tolerance	%	±10	
G	Conformity and Standards			
	General and safety requirements for UPS used in operator access areas		IEC 62040-1	
	Electromagnetic compatibility (EMC) requirements for UPS		IEC 62040-2 (Class C3)	
	Method of specifying the performance and test requirements of UPS		IEC 62040-3 (VFI SS 111)	
	Environmental aspects - requirements and reporting		IEC 62040-4	

BOQ				
S.NO.	Description	Quantity	Unit Price	Total Amount
1	Rack Based kVA/kW modular UPS, Suitable to accommodate Maximum of 5 power modules in single cabinet of 300 kW & 10 power modules in single cabinet of 600 kVA			
2	60 kVA/60 kW Intelligent hot swappable power module with DSP controller			
3	Communication Options			
	SNMP & MODBUS			
	Relay Card			
	Software to monitor multiple UPSs at single console			
4	24 X 7 Remote monitoring services (including required hardwire)			
5	Battery & its accessories			
6	UPS to Battery Cable			
7	Installation & Commissioning Charges			